PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

DDH

1. Definition and epidemiology

- **Developmental dysplasia of the hip (DDH):** spectrum from **acetabular dysplasia** (shallow acetabulum) through **subluxation** to **frank dislocation** (dislocatable or dislocated), developing **before or after birth** (hence "developmental", not "congenital").
- **Incidence: instability about 1-2 per 100 newborns at birth** (most resolve), **true dislocation about 1-1.5 per 1,000**.
- **Teratologic dislocation** (separate entity): dislocated **in utero**, **irreducible**, with **neuromuscular syndromes** (**arthrogryposis, myelomeningocele, Larsen syndrome**): worse prognosis, early open reduction.

2. Risk factors ("4 Fs" + others)

- **Female (about 4-6:1; maternal relaxin sensitivity)**, **First-born** (tight uterus), **Family history (about 10-12x risk with an affected sibling or parent)**, **breech presentation ("Frank breech": hips flexed, knees extended: strongest single risk factor)**.
- **Oligohydramnios**, **left hip (about 60%; left occiput anterior position against the maternal sacrum)**, bilateral in about 20%.
- **"Packaging" disorders** (associated conditions from intrauterine crowding): **congenital muscular torticollis**, **metatarsus adductus**, **congenital knee dislocation**, **calcaneovalgus/clubfoot** (association weaker).
- **Postnatal swaddling with hips extended and adducted** (Native American, Japanese cultural practices; **"hip-healthy swaddling" with hips free to flex and abduct**).

- **Ethnicity:** higher in Native Americans and Sami; low in Africans (carrying on the back with hips abducted).

3. Pathoanatomy

- **Obstacles to reduction (acetabular side, in order):**
  - **Intra-articular: inverted/hypertrophied labrum ("limbus")**, **hypertrophied ligamentum teres**, **pulvinar (fibrofatty tissue in the acetabular floor)**, **contracted inferior capsule**, **transverse acetabular ligament** (hypertrophied).
  - **Extra-articular: iliopsoas tendon** (creates an **hourglass constriction of the capsule**), **adductor longus** (contracture).
- **Secondary changes: shallow, anteverted acetabulum**, **increased femoral anteversion**, **coxa valga**, **false acetabulum (in neglected cases)**, **neolimbus** formation.

4. Clinical examination by age

Newborn (up to about 3 months)

- **Barlow test** (**adduction + posterior pressure on the flexed hip: dislocatable hip**: **"Barlow = Bye-bye"**).
- **Ortolani test** (**abduction + elevation of the greater trochanter: reduces a dislocated hip with a "clunk"**: **"Ortolani = Out to In"**).
- **High-pitched clicks** (soft tissue) are not significant.
- **Both tests become negative after about 3 months** as soft tissues tighten (dislocated hips become irreducible and fixed).

Older infant (over 3 months)

- **Limited abduction (most reliable sign: under 60 degrees in flexion or asymmetry)**.
- **Galeazzi (Allis) sign**: **apparent femoral shortening** with hips and knees flexed (unilateral dislocation; **negative in bilateral DDH**).

- **Asymmetric thigh/gluteal skin folds** (unreliable), **Klisic test** (line from the ASIS through the greater trochanter passes below the umbilicus in a dislocated hip).

Walking child

- **Trendelenburg gait**, **limp**, **leg-length discrepancy**, **waddling gait** (bilateral), **hyperlordosis** (bilateral).

5. Imaging

Ultrasound (up to about 4-6 months: before the femoral head ossific nucleus appears)

- **Graf method (static, coronal view):**
  - **Alpha angle** (**bony roof**: angle between the baseline (ilium) and the bony acetabular roof line): **normal 60 degrees or more**.
  - **Beta angle** (**cartilaginous roof**: baseline to the labrum/cartilage roof line): **normal under 55 degrees** (increases as the head displaces).

Graf type	Alpha	Beta	Description	Management
I	**≥60**	<55	**Normal (mature)**	None
IIa	50-59	55-77	**Physiological immaturity (under 3 months)**	**Repeat ultrasound at 6 weeks to 3 months**
IIb	50-59		Delayed ossification (**over 3 months**)	**Pavlik harness**
IIc	**43-49**	<77	**Critical** (stable or unstable)	**Pavlik**
D	43-49	**>77**	**Decentring** (about to dislocate)	**Pavlik**

Graf type	Alpha	Beta	Description	Management
III	**<43**	>77	**Dislocated (cartilage roof pushed up)**	Pavlik/closed reduction
IV	<43	Labrum inverted (pushed down)	**Dislocated (labrum inverted)**	Closed/open reduction

- **Dynamic ultrasound (Harcke)**: stability; **femoral head coverage (Terjesen/Morin): under 50% abnormal**.
- **Selective ultrasound screening** (**UK NIPE**: risk factors (**breech at or after 36 weeks, family history**) or abnormal examination: **ultrasound by 6 weeks**) vs **universal ultrasound screening** (Austria, Germany: more treatment, possibly fewer late cases).

Radiographs (after about 4-6 months, once the ossific nucleus appears)

- **Hilgenreiner line** (horizontal through the triradiate cartilages), **Perkin line** (vertical from the lateral acetabular margin): **normal: femoral head ossific nucleus (or medial metaphyseal beak) in the inferomedial quadrant**.
- **Shenton line** (broken in subluxation/dislocation).
- **Acetabular index** (**normal under about 30 degrees at birth, decreasing to under 20-25 degrees by 2 years**): increased in dysplasia.
- **Delayed appearance of the ossific nucleus** on the affected side.
- **Centre-edge angle of Wiberg (after about 5 years): normal over 20-25 degrees**.
- **Arthrography** (at the time of closed reduction): **medial dye pool (under 5 mm = acceptable concentric reduction)**, inverted limbus,

"rose thorn" sign (normal labrum).

- **MRI/CT after reduction** in spica: confirm reduction (**MRI preferred: no radiation**).

6. Treatment by age

0-6 months: Pavlik harness

- **Dynamic flexion-abduction orthosis**: **hips flexed about 100 degrees** (anterior straps), **abduction by gravity in the "safe zone" (Ramsey)** (posterior straps loose: prevent adduction but do not force abduction).
- **Safe zone (Ramsey):** the arc between **maximum abduction** and the **position of redislocation (adduction)**: **avoid forced abduction over about 60 degrees** (**AVN**).
- **Positioning errors:**
  - **Excessive flexion (over 120 degrees)**: **femoral nerve palsy** (commonest nerve complication; quadriceps weakness: **stop/adjust harness**), **inferior (obturator) dislocation**,
  - **Insufficient flexion**: failure to reduce,
  - **Forced abduction**: **AVN of the femoral head**,
  - **Brachial plexus palsy** (shoulder straps too tight), skin breakdown.
- **Monitor with ultrasound**: **if not reduced by 3-4 weeks, abandon the Pavlik** (**"Pavlik harness disease"**: **erosion of the posterior acetabular wall** from prolonged use with a persistently dislocated (posterior) head).
- **Success: about 90%+ for dysplastic/subluxated hips**, **about 80-85% for dislocated (Ortolani-positive) hips**; **less effective with Graf IV, bilateral dislocations, Barlow-negative irreducible hips, older age (over 7 weeks at initiation), absent Ortolani sign**.
- **Duration:** full-time until stable (usually about **6-12 weeks**) then weaning.

6-18 months: closed reduction + hip spica

- **Under general anaesthesia**, **arthrogram**, **percutaneous adductor tenotomy** (increases the safe zone).
- **Hip spica cast in "human position" (Salter): hip flexion about 100 degrees, abduction about 40-50 degrees (avoid over 60), neutral rotation**: **about 3 months** in spica (change at 6 weeks), then abduction brace.
- **Post-reduction imaging: MRI or CT** (concentricity).
- **Traction before reduction** (historical: to reduce AVN): **not supported by evidence**.
- **Ossific nucleus presence:** does not affect AVN risk significantly in most studies; **delaying reduction until it appears is not recommended**.

Open reduction: indications

- **Failed closed reduction** (**medial dye pool over 5-7 mm**, unstable in the safe zone, **excessive abduction needed**), **age over about 18-24 months** (variable), **teratologic** dislocation.
- **Approaches:**
  - **Medial approach (Ludloff: between adductor brevis and pectineus/iliopsoas; or Ferguson: posterior to adductor brevis): under about 12-18 months**; addresses medial obstacles (**iliopsoas, ligamentum teres, transverse acetabular ligament, inferior capsule**); **cannot perform capsulorrhaphy**; **medial femoral circumflex artery at risk: higher AVN**.
  - **Anterior (anterolateral, Smith-Petersen/bikini) approach**: **over about 12-18 months**; allows **capsulorrhaphy** and **concomitant pelvic osteotomy**; **iliopsoas tenotomy**, **inverted labrum**, pulvinar.

Over 18-24 months to 8 years: open reduction + femoral and/or pelvic osteotomy

- **Femoral shortening osteotomy** (**reduces pressure on the head: lowers AVN and redislocation**) + **varus derotation** if needed.
- **Pelvic osteotomies:**
  - **Redirectional (reshaping the acetabulum: requires a concentrically reduced, congruous hip)**:
  - **Salter innominate osteotomy** (**single cut above the acetabulum from the sciatic notch to the AIIS; hinges on the pubic symphysis**: **anterolateral coverage**; **age 18 months to about 6 years**; **acetabular index correction up to about 10-15 degrees**; open triradiate cartilage),
  - **Triple innominate osteotomy (Steel, Tönnis)** (**older children/adolescents**; ilium, ischium, pubis),
  - **Periacetabular osteotomy (Bernese/Ganz PAO)** (**closed triradiate cartilage: adolescents and adults**; **posterior column preserved**: allows early weight bearing, preserves pelvic ring),
  - **Reshaping (volume-reducing: incomplete osteotomies hinging on the triradiate cartilage; for capacious acetabula; open triradiate cartilage):**
  - **Pemberton (pericapsular)** (**hinges on the triradiate cartilage**; **reduces acetabular volume**; **anterolateral coverage**; **age 18 months to 10 years**),
  - **Dega** (hinges on the triradiate cartilage; **posterolateral coverage**: **neuromuscular dysplasia (cerebral palsy: posterosuperior deficiency)**),
  - **San Diego** (neuromuscular).
  - **Salvage (augmenting with capsular interposition: fibrocartilage metaplasia; for incongruent hips):**
  - **Chiari osteotomy** (**medial displacement of the acetabulum beneath the ilium**: **capsule interposed**),
  - **Shelf procedure (Staheli)** (bone graft extending the lateral acetabulum).

Over 8 years (bilateral) or over about 10 years (unilateral)

- **Reduction often not attempted** for **high bilateral dislocations without a false acetabulum** (**better function than with AVN/failed reduction**); unilateral: still possible in selected cases; **salvage/arthroplasty later**.

7. Complications

- **Avascular necrosis (AVN) of the femoral head**: **the most important iatrogenic complication**; caused by **forced abduction, excessive pressure, medial circumflex artery injury**: **Kalamchi and MacEwen classification**:
  - **Group I: changes in the ossific nucleus only** (best prognosis),
  - **Group II: lateral physeal damage** (**valgus tilt** of the head, **lateral growth arrest**),
  - **Group III: central physeal damage** (short neck, **coxa breva**, trochanteric overgrowth),
  - **Group IV: total head and physeal damage** (worst).
  - (**Bucholz-Ogden classification** also used.)
- **Redislocation**, **residual dysplasia** (needs pelvic osteotomy), **femoral nerve palsy** (Pavlik), **Pavlik disease**, **stiffness**, **early OA** (**the commonest long-term sequela: the leading cause of THA under age 40-50**), **leg-length discrepancy**, **trochanteric overgrowth** (**greater trochanteric epiphysiodesis or advancement**).
- **Residual dysplasia** is followed with serial radiographs to skeletal maturity.

8. Adult sequelae (DDH in adults)

- **Crowe classification (proximal migration of the femoral head)**: **I under 50% subluxation**, **II 50-75%**, **III 75-100%**, **IV over 100% (high dislocation)**.

- **Hartofilakidis: dysplasia**, **low dislocation** (false acetabulum overlapping the true), **high dislocation** (no contact).
- **THA in DDH: cup in the true acetabulum** (low hip centre), **small cups**, **structural autograft (femoral head) for superolateral deficiency**, **subtrochanteric shortening osteotomy (Crowe IV)** (lengthening **over about 4 cm: sciatic nerve palsy risk**), **excessive femoral anteversion (modular/cemented stems)**.

Exam pearl: **Barlow = dislocatable (adduct + push back); Ortolani = dislocated reduces (abduct + lift). After 3 months: limited abduction is the best sign; Galeazzi negative if bilateral. Breech + female + first-born + family history.**

Exam pearl: **Graf: alpha ≥60 = type I; IIa physiological under 3 months; under 43 = dislocated (III/IV). Ultrasound under 4-6 months, radiographs after (Hilgenreiner, Perkin, Shenton, acetabular index).**

Exam pearl: **0-6 months Pavlik (flexion about 100, no forced abduction; excess flexion = femoral nerve palsy; stop at 3-4 weeks if not reduced: Pavlik disease). 6-18 months closed reduction + spica (human position, adductor tenotomy, medial dye pool under 5 mm). Over 18-24 months open reduction + femoral shortening ± pelvic osteotomy. AVN (Kalamchi-MacEwen) from forced abduction.**

Exam pearl: **Salter: hinge at symphysis, open triradiate, congruent hip. Pemberton/Dega: hinge at the triradiate (Dega: CP, posterior coverage). PAO: closed triradiate, posterior column intact. Chiari/shelf: salvage for incongruent hips (capsular metaplasia).**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.